Review plan · for the teacher

Databases and SQL for school records: unit check

Data engineering · Level 3 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- From sheets to a database
- Asking questions: SELECT, WHERE, ORDER BY
- Counts and averages: GROUP BY
- Joining tables for reports

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (20 min): Read each question aloud from the quiz sheet; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask one team to explain why it's right, using an example from the unit.
4. Wrap-up (8 min): Write the question most teams got wrong on the board, and work through it together.

Discussion

- Which school record would you move into a database first?
- Which query would save you the most time?
- Who reads your reports, and what do they need?

Exit ticket

- One query I will write this week, and who will read its answer.

Try it at school tomorrow

Write one real question about your school as a SQL query (on paper is fine), and say who would read its answer and which columns they need.

Quiz sheet · for each team

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Team _____ Date _____

1 **In a school database, where should each class teacher's phone number be stored?**

◯ In every mark row ◯ Once, in the classes table ◯ In every pupil's row

2 **Match each part of SQL to its job.**

SELECT	• • • •	One row per group
WHERE	• • • •	Which groups come back
GROUP BY	• • • •	Which columns come back
HAVING	• • • •	Which rows come back

3 **Put the parts of a query in the order SQL needs them.**

☐ ORDER BY

☐ GROUP BY

☐ SELECT

☐ LIMIT

☐ WHERE

☐ FROM

4 **This should list 6B's pupils, but returns nothing. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT name
2  FROM pupils
3  WHERE class = '6b';
```

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5 **A class has 30 pupils; 2 missed the test, so their marks are NULL. What do COUNT(*) and COUNT(mark) return for the class?**

◯ 28 and 28 ◯ 30 and 30 ◯ 30 and 28

6 **WHERE COUNT(*) > 30 keeps only the classes with more than 30 pupils.**

◯ True ◯ False

7 **After an inner join of pupils and marks, which of these can happen? Choose all that apply.**

- [ ] A pupil with no mark appears with a zero
- [ ] The query shows a warning about missing pupils
- [ ] A pupil with no mark disappears
- [ ] A pupil with two marks appears twice

8 **Fill the gaps: the names of the pupils who have no mark.**

```
SELECT name FROM pupils
 _____ marks ON marks.pupil_id = pupils.id
WHERE mark _____;
```

Word bank: `JOIN` `= NULL` `LEFT JOIN` `IS NULL`

9 **Complete the sentences.**

A _____ key picks out one row of its table. A _____ key points to a row in another table. _____ means empty: nothing was recorded.

Word bank: total · primary · zero · NULL · foreign

10 **What should a marks report for the head teacher show?**

- ◯ Each class's average mark
- ◯ Every pupil's marks, phone and health notes
- ◯ Nothing: marks are private

Answers · for the teacher

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1 **Once, in the classes table**

Store each fact once: the phone belongs to the class.

2 **SELECT — Which columns come back**
WHERE — Which rows come back
GROUP BY — One row per group
HAVING — Which groups come back

SELECT picks columns, WHERE picks rows before grouping, GROUP BY makes the groups, and HAVING picks groups by their summaries.

3 **Top to bottom: 5, 4, 1, 6, 3, 2**

A query always reads: SELECT columns FROM a table WHERE rows match, GROUP BY a column, ORDER BY a column, LIMIT a number of rows.

4 **Line 3:** `WHERE class = '6B';`

= on text is exact, so '6b' doesn't match '6B'.

5 **30 and 28**

COUNT(*) counts every row; COUNT(mark) skips NULLs.

6 **False**

WHERE works on rows before they're counted, so it can't use COUNT.

7 **A pupil with no mark disappears; A pupil with two marks appears twice**

An inner join keeps only matches, so unmatched pupils vanish silently, with no zero and no warning, and each match makes a row, so two marks make two rows.

8

```
SELECT name FROM pupils
LEFT JOIN marks ON marks.pupil_id = pupils.id
WHERE mark IS NULL;
```

LEFT JOIN keeps every pupil, with NULL where there's no mark, and IS NULL picks those rows.

9 **A primary key picks out one row of its table. A foreign key points to a row in another table. NULL means empty: nothing was recorded.**

Primary keys identify rows, foreign keys link tables, and NULL marks a missing value, which is not the same as zero.

10 **Each class's average mark**

The head needs to compare classes, so averages per class answer the question.